

The empirical recurrence for OEIS A305086, recovered from its tail

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Abstract

OEIS A305086 records “Empirical recurrence of order 97” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 162$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 102$. Its last coefficient is $c_{97} = 8192 \neq 0$, so no further tail satisfies anything shorter; any order-97 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A305086 is “Number of $n \times 5$ 0..1 arrays with every element unequal to 1, 2, 3, 6 or 7 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

3, 70, 182, 352, 3384, 15814, 39016, 211725, ...

The entry states:

Empirical recurrence of order 97 (see link above)

As of the “Last modified” line on the live entry (May 25 2018 08:18:35, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 162$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 162$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 97: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 97.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 324$ exact terms from $n = 5$ on, it returns order 97 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{97} c_i a(n-i),$$

c_1	2	c_{26}	−635383474	c_{51}	99618464719	c_{76}	−200755272
c_2	8	c_{27}	−1234353549	c_{52}	−50811875727	c_{77}	−734109688
c_3	69	c_{28}	−250328188	c_{53}	−141222688329	c_{78}	695748
c_4	−56	c_{29}	2643527510	c_{54}	−88877140164	c_{79}	294742358
c_5	−633	c_{30}	5186102194	c_{55}	33468061133	c_{80}	4216327
c_6	−2239	c_{31}	2408258675	c_{56}	110547399495	c_{81}	−241973758
c_7	550	c_{32}	−6425464261	c_{57}	83298643140	c_{82}	−170028024
c_8	16363	c_{33}	−13508427047	c_{58}	−11272513079	c_{83}	24456196
c_9	43930	c_{34}	−9181285668	c_{59}	−89953240050	c_{84}	89113286
c_{10}	7031	c_{35}	8361261255	c_{60}	−84331668753	c_{85}	45000546
c_{11}	−220783	c_{36}	24119625227	c_{61}	−1218661834	c_{86}	−10612316
c_{12}	−550102	c_{37}	23385530017	c_{62}	76386182879	c_{87}	−15917976
c_{13}	−230335	c_{38}	260888899	c_{63}	74555813524	c_{88}	−2743036
c_{14}	1786544	c_{39}	−27470348314	c_{64}	2645925630	c_{89}	5138912
c_{15}	4359980	c_{40}	−41623308406	c_{65}	−57512562070	c_{90}	2327432
c_{16}	2622643	c_{41}	−28195348839	c_{66}	−51709355974	c_{91}	−586000
c_{17}	−8605536	c_{42}	7587616593	c_{67}	−854702622	c_{92}	−1213504
c_{18}	−20916263	c_{43}	53557060013	c_{68}	34339250112	c_{93}	−435392
c_{19}	−17325303	c_{44}	76669129666	c_{69}	27517541790	c_{94}	−13824
c_{20}	17604406	c_{45}	39528459729	c_{70}	179520862	c_{95}	80384
c_{21}	46335747	c_{46}	−56106979626	c_{71}	−15182758508	c_{96}	32256
c_{22}	68839823	c_{47}	−129566966222	c_{72}	−10688954060	c_{97}	8192
c_{23}	60297940	c_{48}	−86035566529	c_{73}	−33573160		
c_{24}	98169731	c_{49}	55822731307	c_{74}	4540966271		
c_{25}	−120946681	c_{50}	155635073362	c_{75}	2512774797		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 102$, and it is the recurrence OEIS A305086 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{97} - c_1x^{97-1} - \dots - c_{97}$. Its constant term is $-c_{97} = -8192$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 97 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 97, so $\deg q = 97$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 97, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 22 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 97, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 8192.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-97 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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